

METRICS AND APPROXIMABILITY FOR TRIANGULATED CATEGORIES

1. OVERVIEW

Triangulated categories are ubiquitous in algebra, topology, and geometry. Neeman has recently introduced certain metric/analytic techniques for triangulated categories in a series of articles [Nee21b, Nee18, Nee26, BNP23, Nee24, CHNS24, Nee25]¹, and used them to settle multiple conjectures concerning triangulated categories arising in algebraic geometry, and prove other new results².

The idea of this mini-course is to give a feeling of these techniques in action, by going through some of the interesting results proven using them. Most of the applications so far are in algebraic geometry, although these techniques are slowly being applied also in representation theory and homotopy theory, see for example [CG24, Mat26a, Mat26b, LS25].

The first half of the course will be on the specific results proven by this theory. We will postpone many of the details related to metric techniques to the later part of the course, and focus on convincing the listener that there are interesting problems that can be tackled using these methods. In particular, we will look at the connection between the regularity of a Noetherian scheme with the existence of bounded t-structures and strong generators for the category of perfect complexes. Further, we will discuss some new representability theorems and how to use them to give a unified proof of GAGA theorems in algebraic geometry.

In the second half, we will begin by proving that the derived category $D_{\text{Qcoh}}(X)$ of a Noetherian separated scheme is approximable. We will then move on to the passage between various triangulated subcategories of a weakly approximable triangulated category, which is a vast generalisation of a result by Rickard on derived Morita theory. Finally, we will end with a result on the bijection of semiorthogonal decompositions on various (small) triangulated categories associated to a scheme.

Prerequisites. We will assume familiarity with the basics of triangulated categories, or their enhanced incarnations (stable derivators/model/ $(\infty, 1)$ -categories, pretriangulated DG-categories ...). As most of the applications we will discuss are algebro-geometric, we will expect some familiarity with the language of schemes, and with the derived categories and functors associated to them.

2. COURSE PLAN

Week 1: Bounded t-structures and regularity. We will discuss Neeman's resolution of the conjecture by Antieau, Gepner, and Heller [AGH19, Conjecture 1.5] on bounded t-structures implying the regularity of a Noetherian finite dimensional scheme. We will follow [Nee24] with some inputs from [BCM⁺23]. In particular, we will show where weak approximability and preferred equivalence class come in. We will define (weakly) approximable triangulated

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¹In the order the preprints appeared on arXiv.

²See also the survey articles [Nee20, Nee21a, Nee23, CNS25].

categories and, if time permits, prove that the derived categories of connective DG-algebras, ring spectras etc. are approximable.

Week 2: Strong generators and regularity. We will discuss the resolution by Neeman of a conjecture on the relation between existence of strong generators and regularity by Bondal and Van den Bergh [BVdB03]. We will follow the argument in [Nee21b]. In particular, we will define and discuss the proof of compact boundedness, and where approximability comes in.

Week 3: Representability theorems. We will discuss some new representability theorems proven using the metric techniques. We'll begin with the old result by Bondal and Van den Bergh. Then, we'll state Neeman's improvement and use it to give a proof of GAGA. Finally, we'll give a new perspective on how the second part of Neeman's result follows from co-approximability of the mock homotopy category of projective following [MR25b].

Week 4: Approximability for schemes. Following [Nee21b], we will give a proof of approximability for quasi-projective schemes, and then for all separated Noetherian schemes using Chow's lemma.

Week 5. Following [CHNS24], we will prove derived Morita type results on passage between triangulated subcategories of the derived category of a weakly approximable triangulated category. If time permits, we will begin discussing semiorthogonal decomposition and admissible subcategories

Week 6. Following [MR25a], we will prove a result on the bijection of (left/right) admissible subcategories of the various (small) triangulated categories associated to a scheme.

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